Jack Elliott

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EDUCATION

The University of North Carolina at Charlotte - Bachelor of Science in Mechanical Engineering Charlotte, NC | Graduation May 2030

SKILLS

AutoCAD; Fusion360; SolidWorks; Autodesk Eagle; LT Spice; PCB prototyping and assembly; FDM, SLA, & SLS 3D Printing, CAM, Laser Cutting, lean six sigma.

PROJECTS

Electromyography Wearable - ASSIST Center Wearable device challenge North Carolina State University | Feb 2026-Apr 2026

- Researched unmet clinical/fitness needs addressable via wearable EMG sensing and scoped project requirements using a formal engineering design process
- Designed and simulated an analog comparator circuit and ADC signal chain in LTSpice to condition raw EMG signals for digital processing
- Developed, laid out, and assembled a custom low-profile PCB, iterating from prototype to final revision
- Developed an Android application to interface with an ESP-32 over Bluetooth for real-time data transfer and communication
- Fabricated a custom arm sleeve to house and secure electronics for wearable use

Open CV Face tracking delivery Robot - High School Capstone project Cary, NC | Nov 2025 – Jan 2026

- Led a 4-person team through the full engineering design cycle, from concept through manufacturing and testing
- Developed OpenCV-based computer vision software to track faces and generate targeting vectors for a custom 2-axis gimbal
- Designed the complete gimbal and pneumatic air-delivery system assembly in Fusion 360
- Ran FEA simulations, supplemented with AI-assisted optimization, to maximize the gimbal body's strength-to-weight ratio
- Engineered a pneumatic actuation system to drive the delivery mechanism
- Manufactured the gimbal body using FDM 3D printing and CNC milling

EXPERIENCE

Manufacturing Engineering Intern - DotBliss Raleigh, NC | Dec 2024 – Sept 2026

- Contributed to development of 3 novel patented systems related to fabric integrated electronics
- Curated technical data in support of a DARPA-funded program contracted through the U.S. Army Reserve Cyber Operations Group
- Designed and prototyped proprietary mechanical interfaces for fabric integrated electronics
- Developed and designed low-profile PCBs with castellated daughterboards
- Designed and fabricated tool holding systems and jigs for laser cutting processes
- Created custom PCB alignment and assembly jigs to improve manufacturing repeatability

Sales and Service - The Bicycle Chain Apex Apex, NC | July 2024 – August 2025

- Built strong customer relationships by identifying customer needs and recommending tailored products, services, and technical solutions
- Managed inventory and parts ordering to maintain accurate stock levels and ensure timely availability of components
- Diagnosed technical issues and developed practical solutions to address mechanical and equipment-related problems
- Performed preventive maintenance, repairs, and component installations to ensure safe and reliable bicycle operation